

Supplementary Materials

Please note that additional detailed Supplementary Materials (including full code, case studies, and results data) have been hosted in this separate, dedicated Supplementary Materials repository to facilitate access:

<https://anonymous.4open.science/r/CRAVE-D983>

Section S1: Detailed Statistical Tables

This section provides the comprehensive statistical results of the 2×2 factorial evaluation.

S1.1 LLM-as-a-Judge Results

Table S1 summarizes the LLM-as-a-Judge quantitative results. Tables S2 to S4 detail the main effects across the **AllMean Pool**, while Tables S5 to S7 detail the main effects within the **Top-3 Pool**. For extended statistical results, including interaction effects, simple effects, and raw score distributions, please refer to the provided repository.

Table S1: Quantitative Results: Generative Stability (AllMean) vs. Creative Upper-Bound (Top-3)

Criteria	Judge	Pool	RAF		AVL		Main Effect (p)	
			w/o Critic	w/ Critic	w/o Critic	w/ Critic	Guidance (A)	Critic (B)
Relevance (RE)	Gemma-3	AllMean Top-3	4.95	4.91	4.92	4.95	0.749	0.916
			5.00	4.98	4.98	5.00	1.000	1.000
	Mistral-7B	AllMean Top-3	4.45	4.52	4.43	4.52	0.951	0.068
			4.71	4.64	4.81	4.87	0.008	0.833
	Qwen-3	AllMean Top-3	5.00	5.00	5.00	5.00	1.000	1.000
			5.00	5.00	5.00	5.00	1.000	1.000
Coherence (CH)	Gemma-3	AllMean Top-3	4.95	4.94	4.84	4.87	0.001 [†]	0.895
			4.96	4.93	4.87	4.87	0.061	0.641
	Mistral-7B	AllMean Top-3	4.61	4.50	4.72	4.79	0.981	0.421
			4.98	4.72	4.56	4.78	0.013 [†]	0.289
	Qwen-3	AllMean Top-3	5.00	5.00	5.00	5.00	1.000	1.000
			5.00	5.00	5.00	5.00	1.000	1.000
Empathy (EM)	Gemma-3	AllMean Top-3	4.10	4.18	4.35	4.25	<0.001	0.803
			4.20	4.33	4.59	4.43	<0.001	0.782
	Mistral-7B	AllMean Top-3	4.04	4.02	4.11	4.06	0.001	0.039 *
			4.10	4.05	4.24	4.13	0.005	0.018 *
	Qwen-3	AllMean Top-3	5.00	5.00	5.00	5.00	1.000	1.000
			5.00	5.00	5.00	5.00	1.000	1.000
Surprise (SU)	Gemma-3	AllMean Top-3	3.00	3.02	3.05	3.05	0.021	0.581
			3.02	3.07	3.15	3.13	0.019	0.641
	Mistral-7B	AllMean Top-3	3.32	3.34	3.35	3.35	0.524	0.906
			3.66	3.56	3.78	3.73	0.024	0.346
	Qwen-3	AllMean Top-3	2.99	2.98	2.97	2.96	0.597	0.864
			3.15	3.09	3.16	3.11	0.725	0.266
Engagement (EG)	Gemma-3	AllMean Top-3	4.41	4.47	4.57	4.49	0.034	0.803
			4.69	4.70	4.85	4.80	0.026	0.752
	Mistral-7B	AllMean Top-3	4.17	4.20	4.23	4.25	0.043	0.613
			4.39	4.39	4.54	4.54	0.020	0.970
	Qwen-3	AllMean Top-3	5.00	5.00	4.96	4.98	0.007 [†]	0.553
			5.00	5.00	4.98	5.00	0.322	0.322
Complexity (CX)	Gemma-3	AllMean Top-3	3.97	3.96	4.01	3.99	0.087	0.543
			4.04	4.02	4.04	4.06	0.474	1.000
	Mistral-7B	AllMean Top-3	4.02	4.01	4.03	4.01	0.443	0.421
			4.07	4.02	4.09	4.07	0.210	0.210
	Qwen-3	AllMean Top-3	4.05	4.08	3.98	3.99	0.007 [†]	0.507
			4.13	4.19	4.10	4.11	0.290	0.534
Overall (OV)	Gemma-3	AllMean Top-3	4.23	4.24	4.29	4.27	0.010	0.953
			4.32	4.34	4.41	4.38	0.001	0.703
	Mistral-7B	AllMean Top-3	4.15	4.17	4.18	4.19	0.200	0.357
			4.32	4.27	4.39	4.38	0.002	0.478
	Qwen-3	AllMean Top-3	4.51	4.51	4.48	4.49	0.055	0.582
			4.55	4.55	4.54	4.54	0.798	0.519

Note: **Forest green** with **bold** font indicates significant main effects ($p < 0.05$). [†] indicates RAF > AVL. * indicates w/o Critic > w/ Critic. Evaluations from Qwen-3 exhibit a ceiling effect across several criteria and occasional directional divergence from the consensus of the other judge models in specific criteria (EG and CX).

Table S2: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool**, **Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.004 [-0.081, 0.088]	0.008	32385.5	0.951
	Factor B (Critic: w/o vs w/)	-0.079 [-0.163, 0.005]	-0.164	29623.5	0.068
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	-0.194 [-0.409, 0.022]	-0.160	32314.5	0.981
	Factor B (Critic: w/o vs w/)	0.008 [-0.204, 0.221]	0.007	31641.5	0.421
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.056 [-0.091, -0.022]	-0.280	29561.5	0.001
	Factor B (Critic: w/o vs w/)	0.034 [-0.001, 0.069]	0.168	34097.5	0.039
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.020 [-0.100, 0.061]	-0.042	31403.0	0.524
	Factor B (Critic: w/o vs w/)	-0.012 [-0.092, 0.069]	-0.026	32165.0	0.906
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.060 [-0.128, 0.009]	-0.152	29796.0	0.043
	Factor B (Critic: w/o vs w/)	-0.026 [-0.095, 0.042]	-0.066	31705.0	0.613
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.010 [-0.042, 0.022]	-0.055	31783.0	0.443
	Factor B (Critic: w/o vs w/)	0.010 [-0.021, 0.042]	0.057	32866.5	0.421
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.027 [-0.064, 0.011]	-0.122	30280.0	0.200
	Factor B (Critic: w/o vs w/)	-0.015 [-0.053, 0.023]	-0.068	30881.0	0.357

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w/_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S3: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	-0.007 [-0.051, 0.037]	-0.028	32064.5	0.749
	Factor B (Critic: w/o vs w/)	-0.002 [-0.046, 0.041]	-0.009	32254.0	0.916
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.088 [0.037, 0.139]	0.295	35131.0	0.001
	Factor B (Critic: w/o vs w/)	-0.004 [-0.056, 0.049]	-0.012	32216.0	0.895
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.162 [-0.232, -0.091]	-0.396	27076.0	0.001
	Factor B (Critic: w/o vs w/)	0.009 [-0.063, 0.082]	0.022	32627.5	0.803
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.044 [-0.081, -0.007]	-0.205	30895.0	0.022
	Factor B (Critic: w/o vs w/)	-0.011 [-0.048, 0.027]	-0.049	31993.5	0.581
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.094 [-0.181, -0.007]	-0.189	29253.0	0.034
	Factor B (Critic: w/o vs w/)	0.011 [-0.076, 0.098]	0.022	32687.5	0.804
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.037 [-0.081, 0.006]	-0.152	31119.5	0.087
	Factor B (Critic: w/o vs w/)	0.013 [-0.030, 0.056]	0.054	32748.5	0.543
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.043 [-0.071, -0.014]	-0.263	28301.5	0.010
	Factor B (Critic: w/o vs w/)	0.003 [-0.026, 0.032]	0.018	32423.0	0.953

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w/_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S4: Detailed Main Effects of the 2×2 Factorial Evaluation (**AllMean Pool, Qwen3-14B Judge**). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	32294.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	32330.0	1.000
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	0.020 [-0.047, 0.086]	0.052	32833.5	0.597
	Factor B (Critic: w/o vs w/)	0.006 [-0.060, 0.072]	0.016	32505.0	0.864
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	0.030 [0.009, 0.050]	0.241	33258.0	0.007
	Factor B (Critic: w/o vs w/)	-0.007 [-0.028, 0.015]	-0.053	32117.5	0.553
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	0.082 [0.022, 0.142]	0.237	34848.5	0.007
	Factor B (Critic: w/o vs w/)	-0.019 [-0.079, 0.042]	-0.053	31701.0	0.507
Overall (OV)	Factor A (Guidance: RAF vs AVL)	0.022 [0.006, 0.038]	0.235	34584.0	0.055
	Factor B (Critic: w/o vs w/)	-0.003 [-0.020, 0.013]	-0.034	32989.5	0.582

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S5: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool, Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	-0.167 [-0.269, -0.065]	-0.438	4846	0.008
	Factor B (Critic: w/o vs w/)	0.009 [-0.095, 0.114]	0.024	5753	0.833
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.185 [-0.084, 0.454]	0.185	6419	0.013
	Factor B (Critic: w/o vs w/)	0.019 [-0.252, 0.289]	0.018	5581.5	0.290
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.111 [-0.186, -0.036]	-0.398	4940	0.005
	Factor B (Critic: w/o vs w/)	0.083 [0.008, 0.159]	0.296	6581	0.018
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.147 [-0.264, -0.029]	-0.335	4959	0.024
	Factor B (Critic: w/o vs w/)	0.073 [-0.046, 0.191]	0.164	6198	0.346
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.148 [-0.275, -0.022]	-0.314	4872	0.020
	Factor B (Critic: w/o vs w/)	0.000 [-0.128, 0.128]	0.000	5848	0.970
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.037 [-0.098, 0.024]	-0.163	5518.0	0.211
	Factor B (Critic: w/o vs w/)	0.037 [-0.024, 0.098]	0.163	6146.0	0.211
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.092 [-0.144, -0.040]	-0.477	4471	0.002
	Factor B (Critic: w/o vs w/)	0.030 [-0.023, 0.083]	0.153	6148.5	0.478

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w/_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S6: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [-0.026, 0.026]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [-0.026, 0.026]	0.000	5832.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.074 [-0.004, 0.152]	0.257	6264.0	0.061
	Factor B (Critic: w/o vs w/)	0.019 [-0.060, 0.097]	0.064	5940.0	0.641
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	-0.241 [-0.368, -0.113]	-0.507	4428.0	< 0.001
	Factor B (Critic: w/o vs w/)	0.019 [-0.113, 0.150]	0.038	5940.0	0.782
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.093 [-0.170, -0.015]	-0.322	5292.0	0.019
	Factor B (Critic: w/o vs w/)	-0.019 [-0.097, 0.060]	-0.064	5724.0	0.641
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	-0.130 [-0.244, -0.016]	-0.305	5076.0	0.026
	Factor B (Critic: w/o vs w/)	0.019 [-0.097, 0.134]	0.043	5940.0	0.752
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	-0.019 [-0.069, 0.032]	-0.098	5724.0	0.474
	Factor B (Critic: w/o vs w/)	0.000 [-0.051, 0.051]	0.000	5832.0	1.000
Overall (OV)	Factor A (Guidance: RAF vs AVL)	-0.068 [-0.105, -0.031]	-0.489	4469.5	0.001
	Factor B (Critic: w/o vs w/)	0.006 [-0.032, 0.045]	0.043	5991.0	0.703

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w/_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

Table S7: Detailed Main Effects of the 2×2 Factorial Evaluation (**Top-3 Pool, Qwen3-14B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant p -values ($p < 0.05$) are highlighted in bold.

Evaluation Metric	Analyzed Effect	Diff. [95% CI]	Cohen's d	Statistic (U)	p -value
Relevance (RE)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Coherence (CH)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Emotion (EM)	Factor A (Guidance: RAF vs AVL)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
	Factor B (Critic: w/o vs w/)	0.000 [0.000, 0.000]	0.000	5832.0	1.000
Suspense (SU)	Factor A (Guidance: RAF vs AVL)	-0.015 [-0.112, 0.081]	-0.043	5732.0	0.725
	Factor B (Critic: w/o vs w/)	0.053 [-0.044, 0.149]	0.146	6147.0	0.266
Engagement (EG)	Factor A (Guidance: RAF vs AVL)	0.009 [-0.009, 0.028]	0.136	5886.0	0.322
	Factor B (Critic: w/o vs w/)	-0.009 [-0.028, 0.009]	-0.136	5778.0	0.322
Complexity (CX)	Factor A (Guidance: RAF vs AVL)	0.049 [-0.056, 0.155]	0.125	6154.0	0.290
	Factor B (Critic: w/o vs w/)	-0.031 [-0.137, 0.075]	-0.078	5642.5	0.534
Overall (OV)	Factor A (Guidance: RAF vs AVL)	0.007 [-0.015, 0.029]	0.087	5922.0	0.798
	Factor B (Critic: w/o vs w/)	0.002 [-0.020, 0.024]	0.025	6058.5	0.519

Note: Mean Difference for Factor A = $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$. Mean Difference for Factor B = $\text{Mean}_{\text{w/o_Critic}} - \text{Mean}_{\text{w_Critic}}$. [95% CI] represents the 95% confidence interval of the mean difference.

S1.2 Detailed Affective Salience and Intensity Statistics

Table S8 presents the statistical testing on Affective Salience (AS) and the Proportion of High-Salience (HSP). To ensure statistical rigor, the choice between Welch’s t-test and the Mann-Whitney U test was determined dynamically based on the Shapiro-Wilk test for normality.

Table S8: Detailed Statistical Tests on Affective Salience and Intensity. Significance levels: ns ($p \geq 0.05$), * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Metric	Comparison	Mean A	Mean B	Diff. [95% CI]	Cohen’s d	Test Used	p -value
Valence Affective Salience (AS)	RAF_w/o_Critic vs RAF_w/_Critic	0.100	0.096	0.004 [0.000, 0.008]	0.269	Mann-Whitney U	0.040 (*)
	AVL_w/o_Critic vs AVL_w/_Critic	0.107	0.104	0.003 [-0.001, 0.007]	0.211	Welch’s t	0.085 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	0.100	0.107	-0.007 [-0.011, -0.003]	-0.428	Mann-Whitney U	0.003 (**)
	RAF_w/_Critic vs AVL_w/_Critic	0.096	0.104	-0.008 [-0.011, -0.004]	-0.497	Welch’s t	< 0.001 (***)
Arousal Affective Salience (AS)	RAF_w/o_Critic vs RAF_w/_Critic	0.235	0.236	-0.002 [-0.004, 0.001]	-0.148	Welch’s t	0.258 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	0.234	0.235	-0.000 [-0.003, 0.002]	-0.041	Mann-Whitney U	0.895 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	0.235	0.234	0.001 [-0.002, 0.003]	0.071	Mann-Whitney U	0.752 (ns)
	RAF_w/_Critic vs AVL_w/_Critic	0.236	0.235	0.002 [-0.001, 0.005]	0.167	Welch’s t	0.197 (ns)
Valence High-Salience Proportion (%)	RAF_w/o_Critic vs RAF_w/_Critic	23.32	21.85	1.47 [0.07, 2.87]	0.268	Mann-Whitney U	0.057 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	25.81	24.41	1.39 [-0.18, 2.97]	0.213	Mann-Whitney U	0.093 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	23.32	25.81	-2.48 [-3.95, -1.02]	-0.408	Mann-Whitney U	0.004 (**)
	RAF_w/_Critic vs AVL_w/_Critic	21.85	24.41	-2.56 [-4.07, -1.05]	-0.423	Welch’s t	< 0.001 (***)
Arousal High-Salience Proportion (%)	RAF_w/o_Critic vs RAF_w/_Critic	82.65	82.80	-0.16 [-1.11, 0.80]	-0.042	Welch’s t	0.751 (ns)
	AVL_w/o_Critic vs AVL_w/_Critic	82.16	82.55	-0.39 [-1.29, 0.52]	-0.103	Mann-Whitney U	0.417 (ns)
	RAF_w/o_Critic vs AVL_w/o_Critic	82.65	82.16	0.49 [-0.35, 1.33]	0.141	Mann-Whitney U	0.432 (ns)
	RAF_w/_Critic vs AVL_w/_Critic	82.80	82.55	0.26 [-0.76, 1.27]	0.064	Mann-Whitney U	0.754 (ns)

*Note:*The statistical test (Welch’s t or Mann-Whitney U) was selected based on the outcome of the Shapiro-Wilk test for normality on the respective distributions. [95% CI] represents the 95% confidence interval of the mean difference.

Section S2: Prompt Engineering

This section shows the condensed key instructions in the outline and scripts generation prompts. The complete system and user prompts used in the **CRAVE** framework are highly structured. To ensure the readability of this document while maintaining full experimental transparency, we provide:

1. **Core Logic Templates:** The fundamental instructional blocks for the two main mechanisms.
2. **Full Repository:** The raw prompt files, source code, and additional qualitative samples are available at the supplementary repository.

S2.1 AVL Prompts

Figure S1 shows a condensed representation of the core AVL prompts in the outline and scripts generation phases.

S2.2 Critic Prompts

Figure S2 shows a condensed representation of the core Critic evaluation prompts in the outline and scripts generation phases.

Full Dual-Phase AVL Execution Prompt

Core Instruction:

You will be provided with [Valence] and [Arousal] coordinates. EXECUTE this state strictly.

1. Valence Mapping (Plot Direction & Emotional Polarity):

- *Level 1 (Very Negative)* → **Outline:** Major setbacks, tragedies. **Script:** Lexical despair, terror.
- *Level 2 (Negative)* → **Outline:** Minor conflicts, obstacles. **Script:** Tension, sadness, anxiety.
- *Level 3 (Neutral)* → **Outline:** Transitions, balanced events. **Script:** Calm, neutral tone.
- *Level 4 (Positive)* → **Outline:** Small victories, progress. **Script:** Relief, hope, optimism.
- *Level 5 (Very Positive)* → **Outline:** Major resolutions, catharsis. **Script:** Lexical triumph.

2. Arousal Mapping (Event Density & Emotional Intensity):

- *Level 1 (Very Low)* → **Outline:** Reflective moments. **Script:** Linguistic silence, stagnation.
- *Level 2 (Low)* → **Outline:** Slow buildup. **Script:** Introspective, measured pacing.
- *Level 3 (Mid)* → **Outline:** Steady progression. **Script:** Normal conversational pacing.
- *Level 4 (High)* → **Outline:** Fast developments. **Script:** Urgency, increased pressure.
- *Level 5 (Very High)* → **Outline:** Climactic action, twists. **Script:** Chaotic pacing, panic.

Execution Target:

Outlining: Embed the tone into the causal narrative structure.

Scripting: Adopt the state through specific word choice and pacing.

Figure S1: The explicit 5×5 affective constraints provided to the LLM. The mapping dictates macro-level plot events during outlining and micro-level stylistic choices during script generation. The complete prompts for both outlining and scripting stages are available in the Supplementary Material repository.

Simplified Critic Evaluation Prompt

Role: You are an expert evaluation critic judging visual novel generation drafts. Do NOT rewrite; strictly evaluate based on the following dimensions:

1. Intrinsic Narrative Quality:

Ensure the text is engaging, logical, and character-driven within its [GENRE] and [THEME]. It must not be an empty vehicle for emotions.

2. Affective Execution & Progression:

Treat [Valence] (emotional tone) and [Arousal] (pacing/energy) as strict narrative signals.

- *Valence check:* Lexical sentiment and plot setbacks must match the target.
- *Arousal check:* Linguistic urgency, action frequency, and structural density must match the target.

3. Narrative Justification:

The required emotional and pacing shifts must feel meticulously controlled and fully justified by actual narrative events.

[PENALIZE]:

- Emotional shifts that feel forced or unearned.
- Contradictions between tone and pacing (e.g., calm tone with chaotic arousal).

Figure S2: A condensed representation of the core evaluation instructions provided to the Critic Agent. The complete evaluation prompts for both outlining and scripting stages are available in the Supplementary Material repository.

Evaluation Prompt

Theme: {{THEME}}; Genre: {{GENRE}}

Target Story: {{STORY}}

Rate the visual novel story on a scale from 1 to 5 on the following criteria and explain your answer:

1. **Relevance** (how well the story matches its Theme and Genre)
2. **Coherence** (how much the story makes sense)
3. **Empathy** (how well the reader understood the character's emotions)
4. **Surprise** (how surprising the end of story was)
5. **Engagement** (how much the reader engaged with the story)
6. **Complexity** (how elaborate the story is)

Output the ratings and explanations strictly in JSON format.

Structure your response as follows (use these exact keys):

```
{
  "RE_Score": 1-5,
  "RE_Explanation": "Why...",
  "CH_Score": 1-5,
  "CH_Explanation": "Why...",
  "EM_Score": 1-5,
  "EM_Explanation": "Why...",
  "SU_Score": 1-5,
  "SU_Explanation": "Why...",
  "EG_Score": 1-5,
  "EG_Explanation": "Why...",
  "CX_Score": 1-5,
  "CX_Explanation": "Why..."
}
```

Figure S3: Evaluation prompt used in LLM-as-a-Judge Evaluation

Section S3: LLM-as-a-Judge Evaluation Criteria

As introduced in **Section IV-B** of the main manuscript, we employ an LLM-as-a-Judge methodology to quantitatively assess the qualitative aspects of the generated visual novel branches. Figure S3 details the exact prompt utilized to evaluate the narrative across the six distinct dimensions presented in our results. The model is forced to output both a numerical score and a reasoning explanation to ensure the interpretability of the evaluation data.